

Project – Phase 3: SQL Queries and Relational Algebra

In **Phase 3**, you will populate your database and write **SQL queries** on your database. For each query, you will also write the corresponding **Relational Algebra expression**.

Objectives

The goal of this phase is to:

- Practice writing SQL queries on a relational database
 - Use queries involving **selection, projection, joins, sorting, grouping, and aggregate functions**
 - Translate query logic into **Relational Algebra**
 - Understand how SQL and Relational Algebra correspond to each other
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Tasks

1. Populate the Database

Insert enough sample data into your tables so that meaningful queries can be executed.

Requirements:

- Each table must contain at least **5 rows**
 - Foreign key constraints must be respected
 - Your data must support queries involving **joins** and **aggregation**
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2. Write SQL Queries

Each **group member** must write **at least 5 different queries**.

So if a group has:

- 1 members → at least **5 queries**
- 2 members → at least **10 queries**
- 3 members → at least **15 queries**

Each query must include:

- a short description in plain English
- the **SQL query**
- the equivalent **Relational Algebra expression**

Important rules

- Queries written by different members must be **different**
 - Simple repetitions with only small value changes are **not allowed**
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Required Query Categories

Across the whole project, your submitted queries must cover the following categories.

Category A — Basic Queries

1. **Retrieve all rows from a table**
Example: list all students, all products, all customers, etc.
 2. **Selection with WHERE**
Example: list all employees with salary greater than 30000
 3. **Projection of selected columns**
Example: show only customer name and email
 4. **Selection + Projection together**
Example: show names of products with price less than 100
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Category B — Join Queries

1. **Inner join between 2 tables**
Example: list each order with the customer who placed it
2. **Join between 3 tables**
Example: list students, the courses they take, and the instructor of each course
3. **Join with a condition in WHERE**
Example: list orders placed after a certain date with customer names
4. **Join with projection of selected columns**
Example: show employee name and department name

Category C — Aggregate Queries

1. **COUNT**
Example: count number of orders, number of students, number of employees
 2. **COUNT with GROUP BY**
Example: number of orders per customer
 3. **AVG**
Example: average salary, average product price
 4. **MIN**
Example: lowest price, youngest student, earliest order date
 5. **MAX**
Example: highest salary, most expensive product
 6. **SUM** if applicable
Example: total sales amount, total quantity ordered
 7. **Aggregate with GROUP BY and condition**
Example: departments with average salary greater than 50000
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(Optional) Category D — Sorting and Set-Like Query Patterns

1. **ORDER BY ascending**
 2. **ORDER BY descending**
 3. **ORDER BY with multiple columns**
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Category E — Slightly More Complex Queries

1. **Join + Group By**
Example: total number of enrollments per course
2. **Join + Aggregate + Order By**
Example: customers ranked by number of orders
3. **Filtering grouped results with HAVING**
Example: departments with more than 3 employees
4. **Nested query / subquery**
Example: employees with salary greater than average salary
5. **IN / NOT IN query**
Example: students who are enrolled in specific courses

6. EXISTS / NOT EXISTS

Example: customers who have placed at least one order

Minimum Coverage Rules

To ensure that the project includes a variety of query types, your group must select **at least 2 queries from each category above** when designing your queries. Your total number of queries must still satisfy the requirement of **at least 5 queries per group member**.

Format for Each Query

For every query, students must provide the following:

- **Query X – Description**
 - A short English description of what the query does.
 - **SQL script**
 - Write the SQL query.
 - **Relational Algebra Version**
 - Write the equivalent relational algebra expression.
 - Description
 - List the names of students older than 20.
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Deliverables

Students must submit a **zip** file with the name format: **Group05_phase3.zip** where **Group05** represents the group number. The zip file should include:

1. Report (PDF)

The report must include:

- Project title and List of group members
- Brief description of the database
- Queries written by each member
- for each query:

- Description
- SQL version
- Relational algebra version

It should be clear **which member wrote which queries**.

2. SQL Script File (.sql)

The SQL file must include:

- **INSERT** statements used to populate the tables in a **single SQL file** (.sql)
- All SQL queries for this phase in a **single SQL file** (.sql)

The SQL script should run without errors.

3. Appendix — AI Usage Documentation

You are allowed (and encouraged) to use AI tools to assist with your project. However, you must document how AI was used. Please include the following in your report:

- whether you used any AI tools
- the prompts you used
- the AI responses you received
- which parts of the AI output were used directly, modified, or rejected

You may also share a link to your AI conversation instead of copying the full transcript. Make sure the link works.